

The non-mitotic role of HMMR in regulating the localization of TPX2 and the dynamics of microtubules in neurons

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Abstract

A functional nervous system is built upon the proper morphogenesis of neurons to establish the intricate connection between them. The microtubule cytoskeleton is known to play various essential roles in this morphogenetic process. While many microtubule-associated proteins (MAPs) have been demonstrated to participate in neuronal morphogenesis, the function of many more remains to be determined. This study focuses on a MAP called HMMR, which was originally identified as a hyaluronan binding protein and later found to possess microtubule and centrosome binding capacity. HMMR exhibits high abundance on neuronal microtubules and altering the level of HMMR significantly affects the morphology of neurons. Instead of confining to the centrosome(s) like cells in mitosis, HMMR localizes to microtubules along axons and dendrites. Furthermore, transiently expressing HMMR enhances the stability of neuronal microtubules and increases the formation frequency of growing microtubules along the neurites. HMMR regulates the microtubule localization of a non-centrosomal microtubule nucleator TPX2 along the neurite, offering an explanation for how HMMR contributes to the promotion of growing microtubules. This study sheds light on how progenitor cells utilize proteins involved in mitosis for non-mitotic functions.

eLife assessment

In their **valuable** study, Chen et al. investigate the neuronal role of HMMR, a microtubule-associated protein typically associated with cell division. Their findings indicate that HMMR is necessary for proper neuronal morphology and the generation of polymerizing microtubules within neurites, potentially by promoting the function of TPX2. This **solid** body of work is the first step in deciphering the influence of a mitotic microtubule-associated protein in organizing microtubules in neurons and will be of interest to the neurobiology and cytoskeleton fields.

Introduction

Animals interact with the environment through a highly intricate and organized network of interconnected cells. This network, known as the nervous system, is based on cells called neurons that are able to convey signals electrically and chemically. Neurons are also highly polarized with signal outputting compartments called axons that can extend over a meter in length and signal inputting compartments called dendrites that make the most elaborate tree branches pale in comparison. To develop such a complex and polarized morphology, neurons go through a stereotypical morphogenetic process which was initially observed *in vitro* (Dotti et al., 1988 [DOI](#)). Like other cellular processes in which a polarized morphology is established and maintained, neuronal morphogenesis relies on the interplay between different cytoskeletons. The microtubule cytoskeleton in particular is involved in most if not all aspects of the neuronal morphogenetic process. Microtubules are dynamic biopolymers that can undergo polymerization and depolymerization through the addition and removal of α - and β -tubulin heterodimers (Desai and Mitchison, 1997 [DOI](#)). In addition to tubulin heterodimers that make up the bulk of microtubules, a collection of proteins called microtubule-associated proteins (MAPs) also regulate the dynamic nature of this cytoskeleton and play crucial roles in the morphogenesis of neurons (Conde and Caceres, 2009 [DOI](#); Poulain and Sobel, 2010 [DOI](#)). To better understand the role of MAPs in establishing and maintaining the neuronal morphology, we surveyed the microtubule-associated proteome in neurons using affinity purification and quantitative proteomics (Hwang et al., unpublished data). Surprisingly, hyaluronan-mediated motility receptor (HMMR), also known as receptor for hyaluronan mediated motility (RHAMM) or intracellular hyaluronic acid binding protein (IHABP) (Hofmann et al., 1998 [DOI](#)), is among the most abundant MAPs on neuronal microtubules. HMMR was originally identified as a hyaluronan-binding protein from the murine fibroblast (Turley et al., 1987 [DOI](#)). HMMR-targeting antibodies have since been used to demonstrate the involvement of surface-localized HMMR in hyaluronan-dependent motility in a variety of cell types (Hardwick et al., 1992 [DOI](#); Pilarski et al., 1993 [DOI](#); Samuel et al., 1993 [DOI](#); Savani et al., 1995 [DOI](#)). More recent data demonstrate that HMMR is also an intracellular protein and interacts with the microtubule and actin cytoskeletons (Assmann et al., 1999 [DOI](#); Assmann et al., 1998 [DOI](#)). In agreement with its role as a MAP, HMMR has also been shown to interact with other MAPs such as the microtubule motor dynein (Maxwell et al., 2003 [DOI](#)) and the microtubule nucleator TPX2 (Chen et al., 2014 [DOI](#); Groen et al., 2004 [DOI](#); Scrofani et al., 2015 [DOI](#)). Additionally, HMMR plays crucial roles in microtubule assembly near the chromosomes and at the spindle poles, spindle architecture, as well as mitotic progression (Chen et al., 2014 [DOI](#); Groen et al., 2004 [DOI](#); Maxwell et al., 2003 [DOI](#); Scrofani et al., 2015 [DOI](#)). While the association of HMMR with microtubules is well documented in mitotic cells, very little is known about this interaction in non-mitotic cells. *Hmmr* mRNA has been shown to highly express in the proliferative regions of the nervous system in both developing amphibian and murine embryos (Casini et al., 2010 [DOI](#); Li et al., 2017 [DOI](#)). The presence of HMMR protein in the adult brain and in dissociated primary neurons have also been documented (Lindwall et al., 2013 [DOI](#); Nagy et al., 1995 [DOI](#)). Furthermore, HMMR is essential for proper mitotic spindle orientation in neural progenitor cells and the appropriate formation of various brain structures (Connell et al., 2017 [DOI](#); Li et al., 2017 [DOI](#)). These observations demonstrate a crucial mitosis function of HMMR in neural progenitor cells. However, the function of HMMR extends beyond mitosis in neurons. Using antibodies targeting the cell surface HMMR or peptides mimicking hyaluronan binding domain of HMMR, it has been shown that HMMR is involved in neurite extension in primary neurons and intraocular brainstem transplants (Nagy et al., 1995 [DOI](#); Nagy et al., 1998 [DOI](#)). These findings indicate that HMMR plays important roles in the nervous tissue and in non-mitotic neurons. However, whether HMMR exerts any effect on the microtubule cytoskeleton in neurons remains unexplored.

In this study, the function of HMMR in non-mitotic neurons is examined without any preconception regarding its localization. Using the shRNA-mediated depletion, HMMR knockdown negatively impacts the morphology of primary neurons. These morphological phenotypes include the decrease of axon and dendrite length as well as the reduction of axon branching complexity. The opposite phenotypes are observed in neurons transiently expressing *Hmmr*. Both endogenous and exogenous HMMR localize to the microtubule cytoskeleton and exhibit punctate distribution along the neurites. In addition to its microtubule localization, HMMR is observed to enhance the stability and promote the formation of neuronal microtubules. We also found that the effect of HMMR on microtubule formation is due to its role in recruiting the microtubule nucleator TPX2 onto the microtubules. This work demonstrates that HMMR regulates the dynamics of microtubules independent of its mitotic function or its role on the centrosome in non-mitotic cells.

Results

HMMR regulates neuronal morphogenesis

To examine the role of HMMR in non-mitotic neurons, *Hmmr*-targeting shRNA was utilized to knock down *Hmmr* in mouse hippocampal neurons. Hippocampal neurons were selected because they exhibit a high morphological homogeneity, it has been estimated that 85~90% of hippocampal neurons are pyramidal neurons (Banker and Goslin, 1998 [\[4\]](#)). Three different shRNA sequences were used for depleting *Hmmr* in mouse neurons (**Figure S1A**). Upon HMMR depletion, a significant decrease in total neurite length, axon length, dendrite length, and axon branch density can be detected in dissociated hippocampal neurons (**Figure 1A-E** [\[4\]](#)). To eliminate the possibility of the off-target effect, we performed the rescue experiment by co-transfecting plasmids expressing human HMMR (EGFP-hHMMR) and *Hmmr*-targeting shRNA into hippocampal neurons at 0 days *in vitro* (DIV) and incubated for 4 days before fixation and immunofluorescence staining. The expression of EGFP-hHMMR rescues the phenotype of HMMR knockdown (in both total neurite length and axon branching density) (**Figure S1B-E**).

In addition to the loss-of-function assay, we also performed the gain-of-function assay by transiently overexpressing a mouse HMMR fused to AcGFP1 (AcGFP-mHMMR) in hippocampal neurons. Consistent with the knockdown experiments, overexpressing HMMR results in the opposite phenotypes (i.e., an increase in total neurite length, dendrite length, and axon branch density) (**Figure 1F-J** [\[4\]](#)). These functional analyses demonstrate that HMMR plays important roles in regulating the morphogenetic processes in non-mitotic neurons.

HMMR is a microtubule-associated protein in neurons

To understand the cellular mechanism of HMMR in regulating neuronal morphogenesis, the localization of HMMR in neurons was examined. The HMMR antibody was first validated using shRNA-mediated HMMR depletion in neurons. A significant decrease of HMMR immunofluorescence signal was observed in the soma and along the neurites in HMMR-depleted neurons (**Figure S2**), confirming the specificity of the HMMR antibody. Using this antibody, HMMR is detected along the entire neuron with higher abundance in the soma (**Figure 2A** [\[4\]](#)). Given that HMMR is known to interact with microtubules in mitotic cells (Assmann et al., 1999 [\[4\]](#); Maxwell et al., 2005 [\[4\]](#); Tolg et al., 2010 [\[4\]](#)), we examined whether the same interaction exists in neurons.

Hippocampal neurons were permeabilized before fixation (or pre-extracted) to remove cytosolic proteins and to retain proteins anchored to insoluble structures such as cytoskeletal filaments. HMMR signal can be detected along the neurite and in the soma after pre-extraction, indicating that it may be associated with neuronal cytoskeletons (**Figure 2A** [\[4\]](#)). Upon careful examination, we found that both the endogenous HMMR and transiently expressed AcGFP-mHMMR shows punctate localization along the axon and dendrite (**Figure S3**). Furthermore, when transiently overexpressed AcGFP-mHMMR reaches a high abundance level, it colocalizes with microtubules

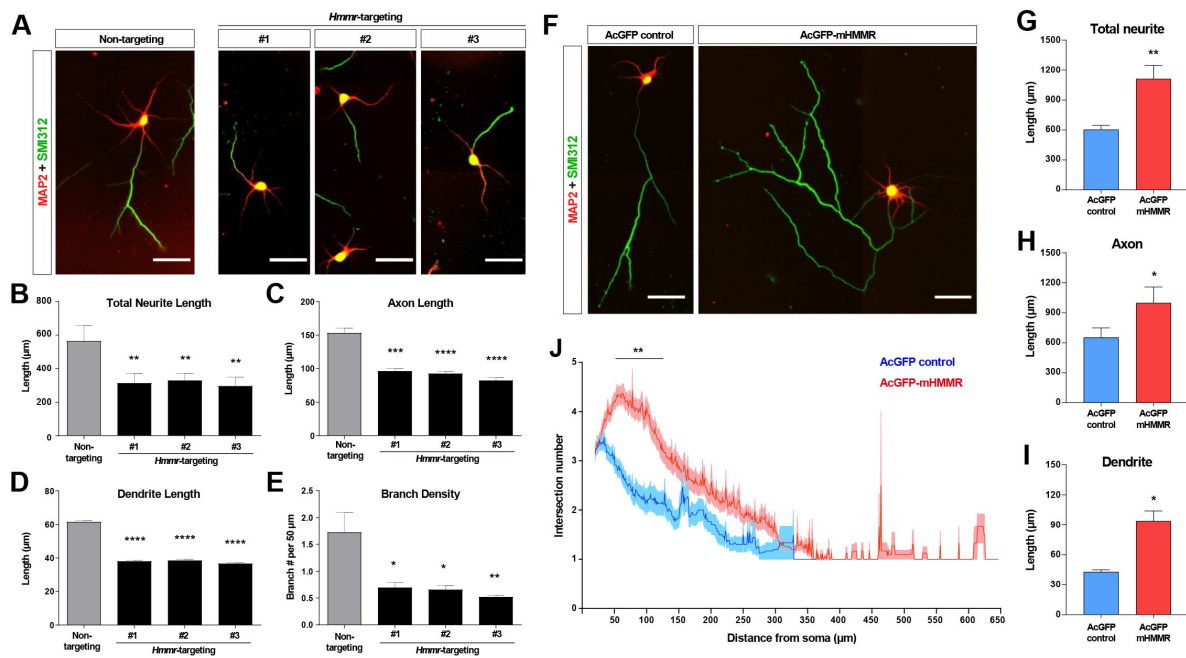


Figure 1.

HMMR promotes neuronal morphogenesis.

(A) Representative images of hippocampal neurons co-transfected with the EGFP-expressing and the indicated shRNA-expressing plasmids on 0 DIV and fixed on 4 DIV. Neurons were immunofluorescence stained with the dendrite marker MAP2 and the axon marker SMI312 (top). Quantification of (B) total neurite length per neuron, (C) axon length, (D) dendrite length, and (E) axon branch density (i.e., branch number per 50 μm of axon). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$, one-way ANOVA followed by Dunnett's post-hoc test. More than 20 neurons were analyzed per condition per repeat. (F) Representative images of hippocampal neurons transfected with AcGFP-or AcGFP-mHMMR-expressing plasmid on 0 DIV and fixed on 3 DIV. Neurons were immunofluorescence stained with the dendrite marker MAP2 and the axon marker SMI312. Quantification of (G) total neurite length per neuron, (H) axon length per neuron, and (I) dendrite length per neuron. * $P < 0.05$, ** $P < 0.01$, two-tailed Student's t-test. More than 50 neurons were analyzed per condition per repeat. (J) Sholl analysis of the axon branching complexity. ** $P < 0.01$, two-way ANOVA followed by Sidak's post-hoc tests. The solid line and shaded area indicate mean and s.e.m. collected from three independent repetitions (more than 50 neurons were analyzed per condition per repetition). All scale bars present 50 μm and all bar graphs are expressed as mean ± s.e.m. from three independent repetitions.

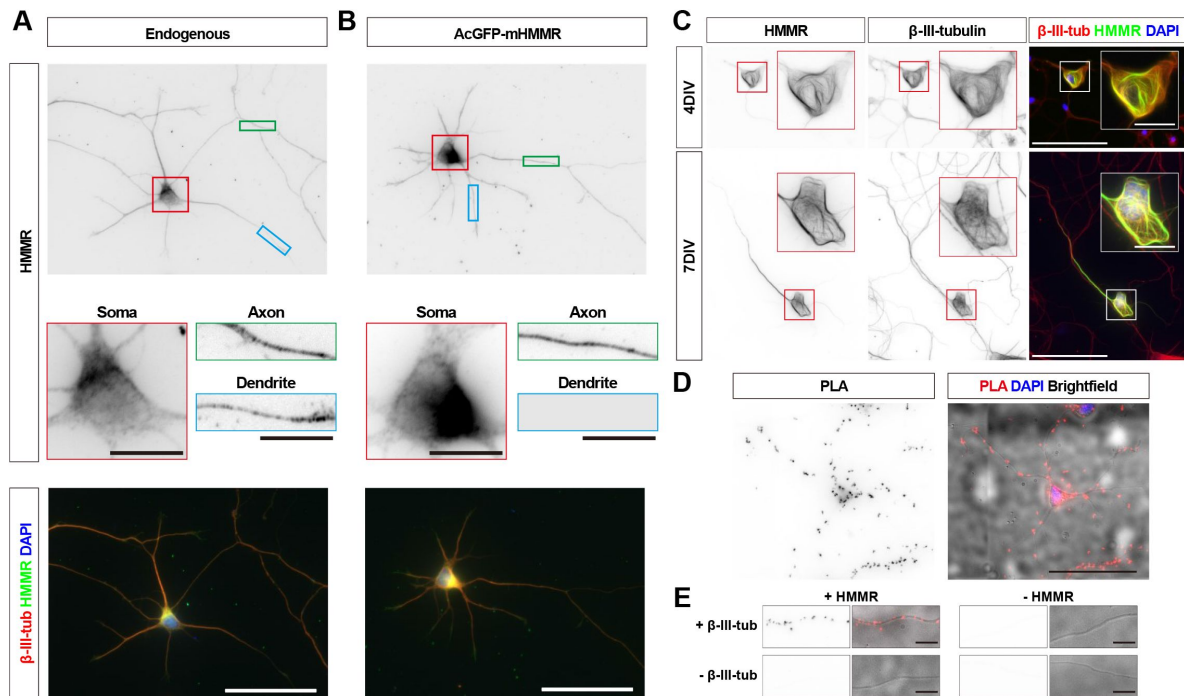


Figure 2.

HMMR localizes to the microtubules in neurons.

(A) Representative images of 3 DIV mouse hippocampal neurons immunofluorescence stained with antibodies against HMMR (green) and β -III-tubulin (red). Nuclei are visualized using DAPI (blue). HMMR images were inverted to improve visualization. (B) Representative images of 3 DIV mouse hippocampal neurons expressing AcGFP-mHMMR. Neurons were fixed and immunofluorescence stained with the antibody against β -III-tubulin (red). Nuclei are visualized using DAPI (blue). AcGFP-mHMMR images were inverted to improve visualization. Colored boxes indicate the magnified regions. The scale bars represent 10 μ m and 50 μ m in the colored boxes and the merged images, respectively. More than 50 neurons were observed for each condition, and HMMR exhibits similar localization in all neurons. (C) Representative images of 4 DIV (top) and 7 DIV (bottom) hippocampal neurons expressing AcGFP-mHMMR. Neurons were immunofluorescence stained with the β -III-tubulin antibody. AcGFP-mHMMR and β -III-tubulin signals were inverted to improve visualization. Red and white boxes at the soma are magnified in the insets. All images have the same scale and the scale bars present 50 μ m. (D) Representative images of proximity ligation assay (PLA) on HMMR and β -III-tubulin in 3 DIV hippocampal neurons. The PLA image was inverted to improve visualization (left). DAPI was used to visualize the nuclei and brightfield microscopy was used to visualize the general appearance of neurons in the merged image (right). The scale bar presents 50 μ m. (E) PLA puncta were present along the neurite shaft only when antibodies against HMMR and β -III-tubulin were both present. All images have the same scale and the scale bars represent 10 μ m.

and sometimes causing the formation of looped microtubules in neurons (**Figure 2B** [↗](#)). To determine whether HMMR is associated with microtubules in neurons, the proximity ligation assay (PLA) was conducted. We first examined whether the endogenous HMMR interacts with microtubules in neurons. Antibodies against HMMR and neuron-specific β -III-tubulin were used in this PLA. If the two primary antibodies are localized in close proximity (<40 nm), an enzymatic reaction will catalyze the amplification of a specific DNA sequence that can then be detected using a red fluorescent probe (Soderberg et al., 2006 [↗](#)).

Consistent with the idea that HMMR associates with neuronal microtubules, fluorescent PLA puncta can be detected in the soma and along the neurite in 3 DIV hippocampal neurons (**Figure 2C** [↗](#)). These fluorescent PLA puncta can only be observed when both primary antibodies were present, indicating that these PLA signals are highly specific (**Figure 2D** [↗](#)). Furthermore, AcGFP-mHMMR-expressing plasmid was transfected into dissociated mouse hippocampal neurons at 0 DIV and incubated for 3 days before fixation and PLA. Antibodies against AcGFP and β -III-tubulin were selected for this PLA. Consistent with the result using endogenous HMMR, fluorescent PLA puncta can be observed in the soma and along the neurite in 3 DIV primary hippocampal neurons (**Figure S4A**). PLA signals can only be observed when both GFP and β -III-tubulin antibodies were present (**Figure S4B**). These results demonstrate that both endogenous and transiently expressed HMMR associate with microtubules in neurons.

Because the punctate distribution of HMMR on microtubules is reminiscent of microtubule plus-ends, we also examined the colocalization of HMMR and EB1 (a microtubule plus-end tracking protein) in neurons. Upon visual examination, HMMR and EB1 do not exhibit colocalization in neurons (**Figure S5A**). To quantify the extent of HMMR and EB1 colocalization in 1 DIV hippocampal neurons, linescans along the neurite and Pearson correlation coefficient were calculated. The average Pearson correlation coefficient is 0.28 ± 0.25 (**Figure S5B**). This result indicates that the punctate HMMR localization in neurons does not represent microtubule plus-ends.

HMMR stabilizes microtubules in neurons

The presence of HMMR on neuronal microtubules and the formation of looped microtubules in AcGFP-mHMMR overexpressing neurons (**Figure 3A** [↗](#)) suggests HMMR may be a microtubule stabilizing factor. To test this possibility, the level of acetylated microtubules was quantified, as this post-translational modification is known to accumulate on stable and long lived microtubules (Schulze et al., 1987 [↗](#)). Consistent with our hypothesis, HMMR depletion via *Hmmr*-targeting shRNA produces a significant decrease of the level of acetylated microtubules in both axons and dendrites (**Figure 3B-D** [↗](#)). In contrast, the level of microtubule acetylation increases in both axons and dendrites of HMMR-overexpressing neurons (**Figure 3E-G** [↗](#)). Taken together, these data demonstrate that HMMR enhances microtubule stability in neurons.

In addition, we examined whether HMMR expression can resist the microtubule destabilizing effect of nocodazole in neurons. AcGFP or AcGFP-mHMMR-expressing plasmid was introduced into dissociated hippocampal neurons on 0 DIV, incubated for 1 day to allow HMMR expression, and administered solvent (DMSO) or 100 nM nocodazole for 2 additional days before neurite length examination. The addition of nocodazole causes a significant reduction of total neurite length, while the expression of HMMR rescues such phenotype (**Figure 3H** [↗](#)). This result is consistent with the idea that HMMR plays a microtubule stabilizing role in neurons.

HMMR regulates neuronal microtubule dynamics

The association of HMMR with neuronal microtubules and its effect on microtubule stabilization suggest that HMMR may also be involved in the regulation of microtubule dynamics in neurons. To investigate this, we utilized the neuronal microtubule dynamics assay previously established in which EB3-mCherry is utilized as a fiduciary marker of growing microtubule plus-ends (Chen et

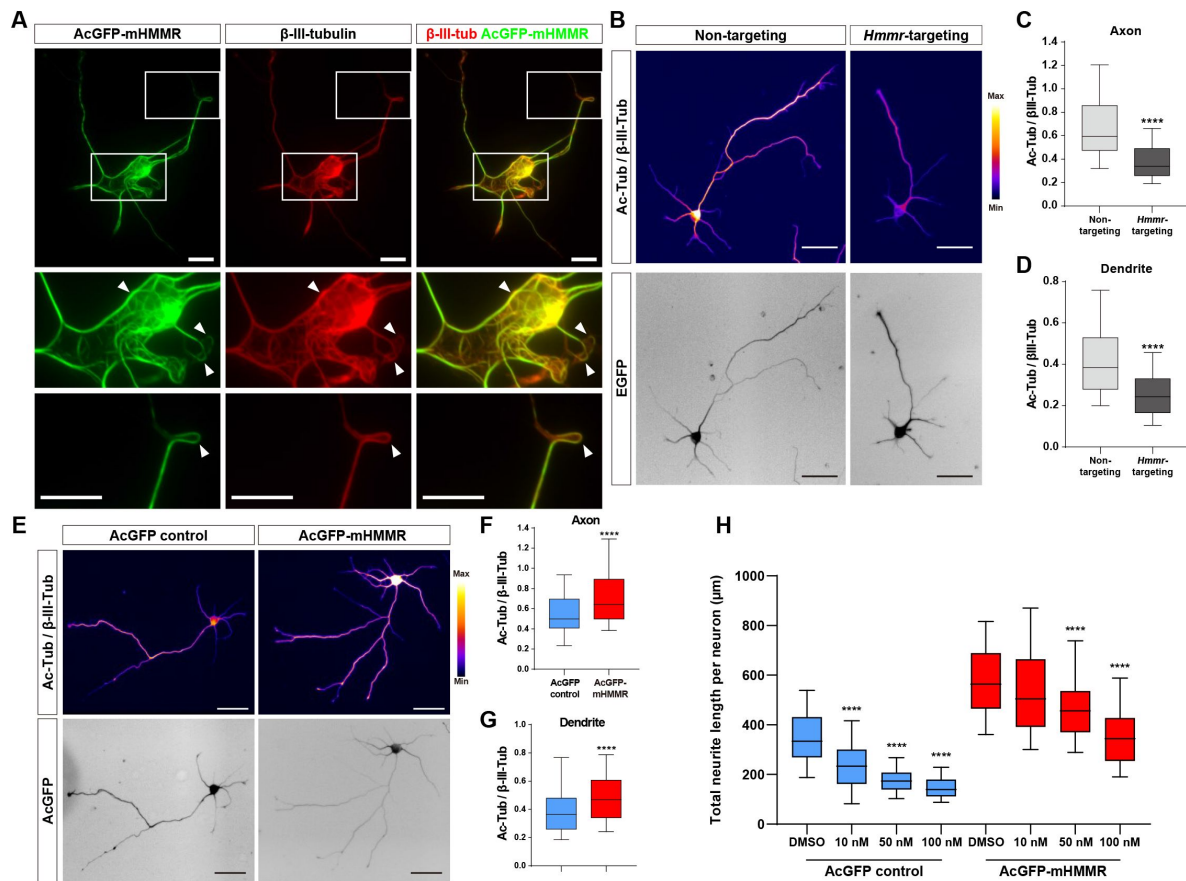


Figure 3.

HMMR regulates microtubule stability in neurons.

(A) Representative images of 4 DIV hippocampal neurons expressing AcGFP-mHMMR. Neurons were immunofluorescence stained with the antibody against β -III-tubulin. White boxes at the soma and the neurite tip are magnified. Arrowheads indicate looped microtubules. (B) Representative pseudo-colored acetylated- α -tubulin-to- β -III-tubulin ratio images of non-targeting shRNA (top left panel) or *Hmnr*-targeting shRNA (top right panel) expressing 4 DIV hippocampal neurons. The transfection indicator EGFP signal was inverted to improve visualization (bottom panels). Only neurons possessing both β -III-tubulin and EGFP signals were quantified. Quantification of the acetylated- α -tubulin-to- β -III-tubulin intensity ratio in axon (C) and dendrite (D). **** $P < 0.0001$, two-tailed Mann-Whitney test. (E) Representative pseudo-colored acetylated- α -tubulin-to- β -III-tubulin ratio images of AcGFP-expressing control (top left panel) or AcGFP-mHMMR (top right panel) expressing 3 DIV hippocampal neurons. The AcGFP signal was inverted to improve visualization (bottom panels). Only neurons possessing both β -III-tubulin and AcGFP signals were quantified. Quantification of the acetylated- α -tubulin-to- β -III-tubulin intensity ratio within axon (F) and dendrite (G). **** $P < 0.0001$, two-tailed Mann-Whitney test. (H) Quantification of total neurite length per neuron in AcGFP-or AcGFP-mHMMR expressing 3 DIV hippocampal neurons treated with or without the indicated concentration of nocodazole for 2 days. **** $P < 0.0001$, one-way ANOVA followed by Dunnett's post-hoc tests comparing to the DMSO groups. All box plots are expressed as first quartile, median, and third quartile with whiskers extending to 5-95 percentile. More than 90 neurons were analyzed per condition per repeat. Scale bars represent 20 μ m in (A) and 50 μ m in (E).

al., 2017 [\[4\]](#)). The rationale is that changes in microtubule dynamics can be detected as alterations in the velocity (or the speed of microtubule polymerization), persistence (the duration of time when EB3-mCherry comet can be followed), and frequency (the number of EB3-mCherry comets detected in a given time span). Plasmids expressing *Hmmr*-targeting shRNA and EB3-mCherry were introduced into dissociated neurons at 0 DIV and incubated for 4 days before fluorescence live cell imaging. To quantify microtubule dynamics, the neurite was separated into 3 different 10 μm regions: proximal, middle, and distal neurite (**Figure 4A** [\[4\]](#)). These 3 regions were selected because of our previous publication (Chen et al., 2017 [\[4\]](#)), in which a significant reduction of EB3 frequency was detected at the tip and the base of the neurite but not in the middle of the neurite in the microtubule nucleator (TPX2) depleted neurons. The reason for this difference is due to the presence of GTP-bound Ran GTPase (RanGTP) at the tip and the base of the neurite. Since RanGTP has been shown to regulate the interaction between HMMR and TPX2 (Scrofani et al., 2015 [\[4\]](#)), it is possible that the same regulation mechanism exists in neurons. HMMR depletion results in a significant decrease of EB3-mCherry emanation frequency in all 3 neurite regions (**Figure 4B-C** [\[4\]](#)). Moreover, a trend of increased microtubule polymerization velocity and a trend of decreased persistence are observed in HMMR-depleted neurons. Next, we examined whether opposite effects on microtubule dynamics can be detected in HMMR overexpressing neurons. Plasmids expressing AcGFP-mHMMR and EB3-mCherry were introduced into dissociated neurons at 0 DIV and incubated for 4 days before fluorescence live cell imaging and microtubule dynamics assay. Congruent with the depletion results, a significant increase in EB3-mCherry emanation frequency is observed at the proximal, middle, and distal neurites in HMMR-expressing neurons (**Figure 4D-E** [\[4\]](#)). Further agreement comes from the decrease of microtubule polymerization velocity and the increase of microtubule persistence in neurons overexpressing HMMR. Both the decrease of microtubule polymerization velocity and the increase of microtubule persistence are consistent with the observation that HMMR can enhance the stability of neuronal microtubules. These data demonstrate that HMMR regulates the dynamics of microtubules in neurons.

HMMR promotes TPX2-microtubule interaction in axons and dendrites

While changes in microtubule polymerization velocity and persistence in the previous section can be explained by the microtubule stabilizing effect of HMMR, the alteration in microtubule emanation frequency cannot. One explanation is that HMMR influences the function of another microtubule dynamics regulator in neurons. It has been shown that HMMR interacts with the microtubule nucleator TPX2 in a cell cycle-dependent manner (Maxwell et al., 2005 [\[4\]](#)) and this interaction is required for concentrating TPX2 at the spindle poles (Groen et al., 2004 [\[4\]](#)). Furthermore, we have shown that microtubule-bound TPX2 localizes along the neurite and is responsible for non-centrosomal microtubule formation in neurons (Chen et al., 2017 [\[4\]](#)). Combining these two observations, we hypothesized that HMMR affects the localization of TPX2 along the neurite which in turn regulates the formation of neuronal microtubules. To examine this hypothesis, the localization of TPX2 along the neurite was examined in neurons with or without HMMR depletion. Since the abundance of TPX2 along the neurite is rather low (Chen et al., 2017 [\[4\]](#)), PLA was utilized to detect this localization of TPX2. PLA using antibodies against TPX2 and β -III-tubulin produces numerous puncta along the neurite (**Figure 5A** [\[4\]](#)). This punctate distribution appears similar to that produced by HMMR and β -III-tubulin PLA (**Figure 2C** [\[4\]](#)). Consistent with our hypothesis, a statistically significant increase of PLA inter-punctal distance (the distance between PLA puncta) is detected in both axons and dendrites of HMMR-depleted neurons (**Figure 5B-C** [\[4\]](#)). This result indicates that HMMR depletion reduces the localization of TPX2 on neuronal microtubules. These fluorescent PLA puncta can only be observed when both primary antibodies were present, indicating that these PLA signals are highly specific (**Figure 5D** [\[4\]](#)). Next, the effect of HMMR overexpression on TPX2 localization is examined. Plasmids expressing AcGFP-mHMMR were introduced into hippocampal neurons on 0 DIV and incubated for 7 days before fixation and PLA. In agreement with the depletion experiment, a significant

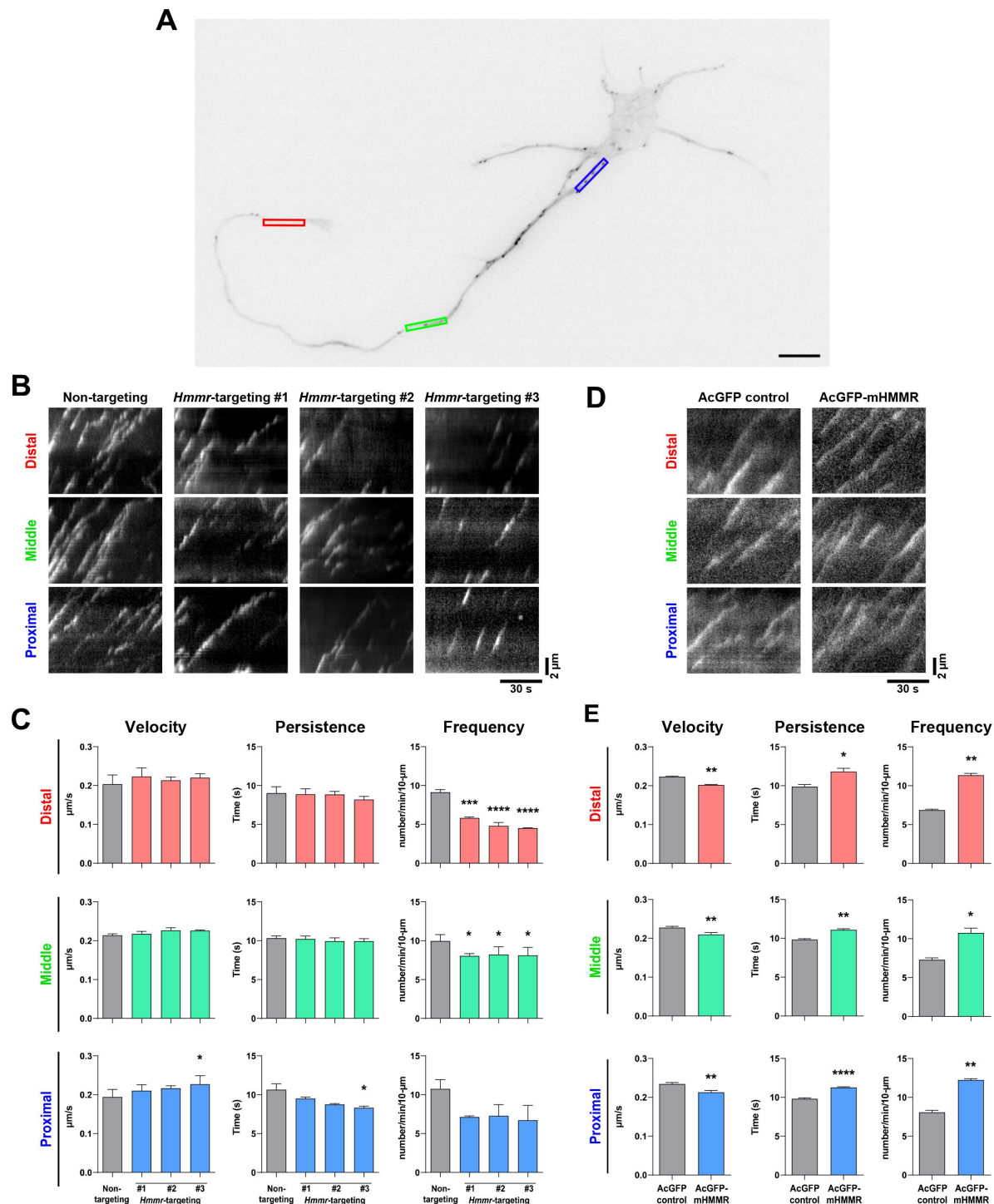


Figure 4.

HMMR regulates the dynamics of neuronal microtubules.

(A) Representative image of a 4DIV EB3-mCherry-expressing cortical neuron. The color boxes indicate regions of quantification: red, green, and blue boxes represent the distal, middle, and proximal neurite, respectively. The scale bar presents 10 μ m. (B) Representative kymographs of indicated neurons at different regions of the neurite. (C) Quantification of EB3-mCherry comets dynamics in B. * $P<0.05$, *** $P<0.001$, **** $P<0.0001$, one-way ANOVA followed by Dunnett's post-hoc tests. (D) Representative kymographs of indicated neurons at different regions of the neurite. (E) Quantification of EB3-mCherry comets dynamics in D. * $P<0.05$, ** $P<0.01$, **** $P<0.0001$, two-tailed Student's t-test. At least 15 neurons were analyzed per condition per repeat. All bar graphs are expressed as mean $\pm$ s.e.m. from three independent repeats.

decrease in PLA inter-punctal distance is observed in HMMR overexpressing axon and dendrite (**Figure 5E-G**). This result indicates that HMMR overexpression enhances the localization of TPX2 on neuronal microtubules. PLA puncta can only be observed when both primary antibodies were present (**Figure 5H**). Taken together, these results demonstrate that HMMR promotes the interaction between TPX2 and microtubules in neurons.

Discussion

In this study, we demonstrate that HMMR influences cellular morphogenesis in non-mitotic neurons using loss- and gain-of-function assays. Transient expression of HMMR promotes axon and dendrite elongation as well as enhances branch density, while depletion of HMMR produces the opposite phenotypes. Both endogenous and transiently expressed HMMR localize primarily to the microtubule cytoskeleton in neurons. The distribution of HMMR along the neurite has a punctate appearance but does not colocalize with the microtubule plus-ends. Using transient expression and shRNA-mediated depletion, it was discovered that HMMR enhances the stability and promotes the formation of neuronal microtubules. Finally, we show that HMMR regulates the recruitment of the microtubule nucleator TPX2 onto neuronal microtubules. These results demonstrate for the first time that HMMR plays an important role in regulating microtubules and morphogenesis in non-mitotic cells.

HMMR has been documented to be a hyaluronate-binding protein (Turley et al., 1987) as well as a microtubule-associated protein (Assmann et al., 1999). In more recent studies, HMMR is found to be associated with the centrosome in mitotic cells (Maxwell et al., 2003). It contains a central rod domain with coiled-coil structures flanked by two microtubule-binding domains at the N-terminus and a centrosome-targeting bZip motif at the C-terminus (Assmann et al., 1999; Maxwell et al., 2003). While numerous studies focus on the role of HMMR in mitotic cells, studies on non-mitotic neurons are scarce. It has been documented that neutralizing HMMR with a functional blocking antibody compromises neurite extension in cell line and primary neurons (Nagy et al., 1995). A similar neutralizing strategy was used to demonstrate that HMMR is involved in axon outgrowth in an intraocular transplantation model (Nagy et al., 1998). Our loss-of-function assay is consistent with these aforementioned publications. It has been shown that the *Xenopus* HMMR homolog XRHAMM bundles microtubules *in vitro* (Groen et al., 2004). In addition, deleting proteins which promote microtubule bundling (e.g., doublecortin knockout, MAP1B/MAP2 double knockout) leads to impaired neurite outgrowth (Bielas et al., 2007; Teng et al., 2001). These observations are consistent with our data that overexpressing HMMR leads to the increased axon and dendrite outgrowth, while depleting it results in the opposite phenotype (**Figure 1**). In addition, our data indicate that the increase of microtubule stability is the main mechanism driving the enhanced neurite outgrowth in neurons overexpressing HMMR (**Figure 3H**). It is worth noting that the elevated level of HMMR increases the branching density of axons (**Figure 1J**) and promotes the formation of looped microtubules (**Figure 3A**). This is consistent with the observations that looped microtubules are often detected in regions of axon branching site prior to branch formation (Dent et al., 1999; Dent and Kalil, 2001; Purro et al., 2008).

Using EB3-mCherry as a marker for growing microtubule plus-ends, we discovered that HMMR increases the amount of growing microtubules. These growing microtubules can either come from the *de novo* formation of microtubules or the repolymerization from paused or shrinking microtubules. The explanation of this increased number of growing microtubules lies in the observation that the localization of TPX2 (a branch microtubule nucleator) on microtubules is regulated by HMMR in neurons. This is consistent with previous publications showing that HMMR interacts with TPX2 and is required for concentrating TPX2 at the spindle pole (Groen et al., 2004; Maxwell et al., 2005; Scrofani et al., 2015). Given that GTP-bound Ran (RanGTP) promotes the interaction between HMMR and TPX2 in the cell-free system (Scrofani et al., 2015), it is tempting to hypothesize that RanGTP also regulates the HMMR-TPX2 interaction in neurons. It

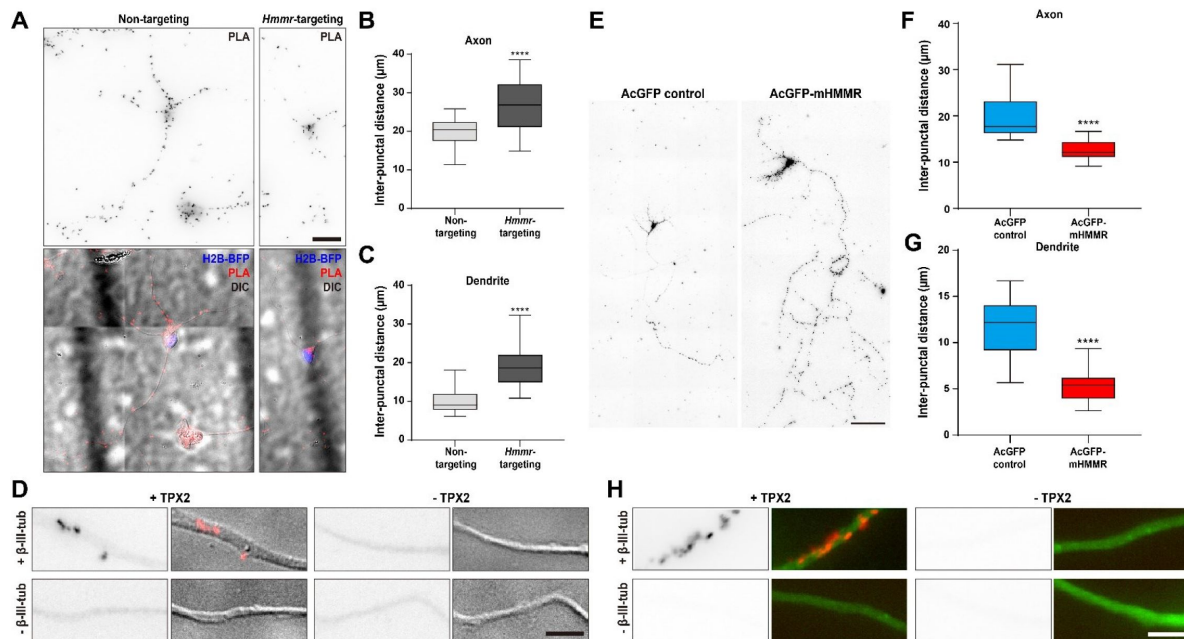


Figure 5.

HMMR regulates the localization of TPX2 on microtubules in neurons.

(A-D) Hippocampal neurons were co-transfected with indicated shRNA- and H2B-BFP-expressing plasmids at 0 DIV and cultured for 4 days before ice-cold methanol fixation. The *Hmnr*-targeting #1 shRNA was utilized to deplete HMMR. (A) Representative PLA images for TPX2 and β-III-tubulin (upper panel) and merged images of PLA, H2B-BFP, differential interference contrast (DIC) (lower panel) in 4 DIV hippocampal neurons. Quantification of the inter-punctal distance of PLA signals in axon (B) and dendrite (C). Only neurons possessing both PLA and H2B-BFP signals were quantified. **** $P < 0.0001$, two-tailed Mann-Whitney test. (D) PLA signals were presented along the neurite only when both TPX2 and β-III-tubulin antibodies were present. (E-H) Hippocampal neurons were co-transfected with H2B-BFP- and either AcGFP- or AcGFP-mHMMR-expressing plasmids at 0 DIV and cultured for 7 days before ice-cold methanol fixation. (E) Representative PLA images for TPX2 and β-III-tubulin in 7 DIV hippocampal neurons. Quantification of the inter-punctal distance of PLA signals in axon (F) and dendrite (G). Only neurons possessing both PLA and H2B-BFP signals were quantified. **** $P < 0.0001$, two-tailed Mann-Whitney test. All box plots are expressed as first quartile, median, and the third quartile with whiskers extending to 5-95 percentile. (H) PLA signals were presented along the neurite only when both TPX2 and β-III-tubulin antibodies were present. Scale bars represent 20 μm in (A), 5 μm in (D) (H), and 100 μm in (E).

has previously been shown that cytoplasmic RanGTP promotes the formation of non-centrosomal microtubules at the neurite tip (Huang et al., 2020 [DOI](#)). This is due to the effect of RanGTP on releasing TPX2 from the inhibitory importin heterodimers (Chen et al., 2017 [DOI](#)). Our results suggest that the effect of cytoplasmic RanGTP on neuronal microtubules may come from recruiting TPX2 to the existing microtubules as well as enhancing its nucleator activity. It has been shown that compromising microtubule nucleation in neurons by SSNA1 mutant overexpression prevents proper axon branching (Basnet et al., 2018 [DOI](#)). Additionally, dendritic branching in *Drosophila* sensory neurons depends on the orientation of microtubule nucleation. Nucleation that results in an anterograde microtubule growth leads to increased branching, while nucleation that results in a retrograde microtubule growth leads to decreased branching (Yalgin et al., 2015 [DOI](#)). These results demonstrate the importance of microtubule nucleation on neurite branching. It is conceivable that overexpressing a microtubule nucleation promoting protein such as HMMR results in an increase of axon branching complexity.

Neurons are the communication units of the nervous system. The formation of their intricate shape is therefore crucial for the physiological function. Alterations in neuronal morphogenesis have a profound impact on how nerve cells communicate, leading to a variety of physiological consequences. These consequences conceivably include impaired neural circuit formation and function, compromised signal transmission between neurons, as well as altered anatomical structure of the CNS. Depending on the specific type and location of the morphogenetically altered neurons, the physiological consequences can include neurological disorders such as autism spectrum disorder (Berkel et al., 2012 [DOI](#)) and schizophrenia (Goo et al., 2023 [DOI](#)), as well as learning and memory deficits (Winkle et al., 2016 [DOI](#)). However, due to the involvement of HMMR on mitosis, most HMMR mutations are associated with familial cancers in humans (based on ClinVar data).

Given the importance of HMMR on spindle integrity and orientation, studies of HMMR on neural development have largely focused on these aspects. It has been documented that neural progenitor cells from *Hmmr* knockout or C-terminal truncation mice exhibit misoriented spindle and consequently these animals develop either megalencephaly or microcephaly (Connell et al., 2017 [DOI](#); Li et al., 2017 [DOI](#)). Interestingly, *Hmmr* depletion in *Xenopus* showed defects in anterior neural tube closure (Prager et al., 2017 [DOI](#)). During CNS development, the anterior part of the neural tube closes and differentiates into the brain while the posterior part becomes the spinal cord. A series of cellular actions take place during the neural tube closure that include polarization, migration, and intercalation (Nikolopoulou et al., 2017 [DOI](#)). This failure in closing the anterior neural tube suggests an underlying defect in the microtubule cytoskeleton. Upon careful examination, interphase cells from the deep neural layer in *Hmmr* depleted embryos adopted a web-like microtubule organization instead of their typical linear microtubule organization. This data is consistent with our observation that HMMR plays a role in organizing the microtubule cytoskeleton in non-mitotic cells of the neural tissue. In support of this idea, *Hmmr* is amongst the highest expressed RNA in the corpus callosum relative to other tissues in adult humans (Rouillard et al., 2016 [DOI](#)). Because HMMR plays such a critical role in mitosis of neural progenitor cells, any loss-of-function *Hmmr* mutation in humans will likely result in embryonic lethality and prevent the observation of non-mitotic, microtubule-based phenotypes such as corpus callosum malformation or lissencephaly. It will be of great interest to examine the microtubule-rich brain structures such as corpus callosum or microtubule-dependent processes such as neuronal migration in *Hmmr* conditional knockout mice.

Materials and Methods

Antibodies and reagents

Acetylated- α -tubulin antibody (ab24610) was purchased from Abcam (Cambridge, United Kingdom). β -III-tubulin antibodies TUJ1 and TUBB3 (801202 and 802001) as well as neurofilament monoclonal antibody SMI312 (837904) were from BioLegend (San Diego, CA). GFP antibody (12A6) was from DSHB (Iowa City, IA). HMMR antibody E-19 (sc-16170) was from Santa Cruz Biotechnology (Dallas, TX). MAP2 antibody (AB5622) and Duolink proximity ligation assay were from MilliporeSigma (Burlington, MA). TPX2 antibody was a kind gift from Oliver Gruss (Gruss et al., 2002 [DOI](#)). Alexa Fluor-conjugated secondary antibodies were from Thermo Fisher Scientific (Waltham, MA).

Plasmids

The mouse *Hmmr*-expressing plasmid pCAG-AcGFP-mHMMR was cloned by inserting wild-type mouse *Hmmr* gene obtained from the mouse embryonal carcinoma P19 cell cDNA using PCR primers (5'-ATAGTCGACAGGCGTCAGAATGTCCTTCCT-3' and 5'-TACCCGGGACTTCCATGATTCTTGAAGTTGCA-3') into pCAG-AcGFP-C3 using SalI and XmaI restriction endonucleases. The *Hmmr* gene obtained is 2,385 bp in length and translates into a protein ~92 kDa in molecular weight. The human *HMMR*-expressing plasmid pEGFP-hHMMR was a kind gift from Dr. Christopher Maxwell (Maxwell et al., 2003 [DOI](#)). The mouse *Hmmr*-targeting shRNA plasmids were obtained from the RNAi Core of Academia Sinica (Taipei, Taiwan).

Neuron culture and transfection

All animal experimental procedures were approved by the Institutional Animal Care and Use Committee (IACUC) and in accordance with the Guide for the Care and Use of Laboratory Animals of National Yang Ming Chiao Tung University. Dissociated hippocampal and cortical neuron cultures were prepared as previously described (Chen et al., 2017 [DOI](#)) with the following modifications. Hippocampi or cortices from E17.5 mouse embryos were dissected, digested with trypsin-EDTA and triturated. Dissociated neurons were seeded onto poly-L-lysine-coated coverslips (2.5×10^3 cells/cm² for low-density cultures and 3×10^4 cells/cm² for regular-density culture). Plasmids were introduced into neurons using Nucleofector II (Lonza, Basel, Switzerland) immediately before seeding or using Lipofectamine 2000 (Thermo Fisher Scientific) at the indicated number of days *in vitro*. Lipofectamine transfected cells were incubated for 4 h and the medium containing the transfection mixture was then replaced with cortical neuron-conditioned neurobasal medium (Thermo Fisher Scientific, 21103049) (low-density cultures) or fresh neurobasal medium plus B27 supplement (regular-density culture).

Indirect immunofluorescence staining

Cells on coverslips were fixed with 3.7% formaldehyde for 15 min at 37 °C and then washed three times with PBS. Fixed cells were permeabilized with 0.25% triton X-100 in PBS for 5 min at room temperature or extracted in -20 °C methanol for 10 min. For experiments that required cytosolic pre-extraction, cells on coverslips were permeabilized in 0.1% triton X-100 in PIPES buffer (0.1 M PIPES pH 6.9, 1 mM MgCl₂, and 1 mM EGTA) for 15 sec, washed once with PIPES buffer, and fixed with 3.7% formaldehyde in PIPES buffer at 37 °C for 30 min and then washed with PBS three times. Cells were then blocked with 10% BSA in PBS for 30 min at 37 °C, incubated for 1 hour at 37 °C with different primary antibodies: GFP (1:100), HMMR (1:50), MAP2 (1:1000), SMI312 (1:1000), TPX2 (1:2000), acetylated- α -tubulin (1:1000), TUBB3 (1:2000), and TUJ1 (1:4000). After primary

antibody incubation, cells were washed with PBS three times and incubated with AlexaFluor-conjugated secondary antibodies (1:1000). All antibodies were diluted in 2% BSA in PBS. Coverslips with cells were washed with PBS three times and mounted with Fluoromount onto glass slides.

In situ proximity ligation assay (PLA)

Cells were fixed in -20 °C methanol for 10 min, washed with PBS, and then blocked in a chamber with Duolink II Blocking Solution for 30 min at 37 °C. Primary antibodies used for different experiments were diluted in PBS containing 2% BSA at aforementioned dilutions and incubated for 1 h at 37 °C. Cells were then incubated with PLA probes diluted in Antibody Diluent for 1 h at 37 °C. Subsequent procedures were conducted according to the manufacturer's instructions.

Microscopy acquisition

Fluorescence images were acquired on a Nikon Eclipse-Ti inverted microscope equipped with a Photometrics CoolSNAP HQ2 CCD camera, an Intensilight epi-fluorescence light source, and Nikon NIS-Element imaging software. 20 × 0.75 N.A. or 60 × 1.49 N.A. Plan Apochromat objective lenses were used to collect fluorescence images.

Live cell imaging was performed on a Nikon Eclipse-Ti inverted microscope equipped with a TIRF illuminator and a Tokai Hit TIZHB live cell chamber. Images were acquired using a 60 × 1.49 N.A. Plan Apochromat objective lens, a 561 nm DPSS laser, a Photometrics CoolSNAP HQ2 camera, and Nikon NIS-Elements imaging software. The built-in perfect focus system (PFS) was activated to maintain axial position. Images were acquired every 500 milliseconds over a 2-minute period. Only the neurons with clear EB3 comets were imaged.

Image analysis

For neurite length analysis, fluorescence images were manually traced with the ImageJ plugin NeuronJ 1.4.1 (Meijering et al., 2004 [DOI](#)). Only neurons expressing both the transfection indicator (e.g., EGFP) and specific markers (e.g., β -III-tubulin, MAP2, SMI312) were analyzed. Only neurites longer than its soma diameter were analyzed. For axon branching analysis, neurites were manually traced using the Fiji plug-in Simple Neurite Tracer (Longair et al., 2011 [DOI](#)) before being processed with the Fiji plug-in Sholl analysis (Ferreira et al., 2014 [DOI](#)).

For acetylated microtubule quantification, manually generated linescans along the neurite were used to obtain the signal of acetylated- α -tubulin and β -III-tubulin. The ratio of acetylated- α -tubulin/ β -III-tubulin was calculated to represent the level of microtubule acetylation.

For microtubule plus-end dynamics analysis, NIS-Elements software was used to generate the kymograph for the EB3-mCherry images. All kymographs were generated using a window 10 μ m in length and 7 pixels in width. For proximal neurite analysis, the kymograph window started from the edge of the soma and extended outwards. For mid-neurite analysis, the kymograph window was centered at the midpoint of the neurite. For distal neurite analysis, the kymograph window started at the wrist of the growth cone and extended inwards. The speed and persistence time of EB3-mCherry were quantified from the kymograph by drawing a line along an EB3-mCherry event. Only EB3-mCherry movements that could be followed clearly for equal or more than four frames (1.5 seconds) were defined as an event. The emanating frequency of EB3-mCherry was quantified from the kymograph by counting the number of EB3-mCherry events per minute.

Statistical analysis

All statistical analyses were performed using GraphPad Prism 8. Significant difference between the means was calculated with the indicated statistical methods.

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Editors

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Reviewer #1 (Public Review):

The microtubule cytoskeleton is essential for basic cell functions, enabling intracellular transport, and establishment of cell polarity and motility. Microtubule-associated proteins (MAPs) contribute to the regulation of microtubule dynamics and stability - mechanisms that are specifically important for the development and physiological function of neurons. Here, the authors aimed to elucidate the neuronal function of the MAP Hmnr, which they had previously identified in a (yet unpublished) quantitative study of the proteome associated with neuronal microtubules. The authors conduct well-controlled experiments to demonstrate the localization of endogenous as well as exogenous Hmnr on microtubules within the soma as well as all neurites of hippocampal neurons. Functional analysis using gain- and loss-of-function approaches demonstrates that Hmnr levels are crucial for neuronal morphogenesis, as the length of both dendrites and axons decreases upon loss of Hmnr and increases upon Hmnr overexpression. In addition to length alterations, the branching pattern of neurites changes with Hmnr levels. To uncover the mechanism of how Hmnr influences neuronal morphology, the authors follow the lead that Hmnr overexpression induces looped microtubules in the soma, indicative of an increase in microtubule stability. Microtubule acetylation indeed decreases and increases with Hmnr LOF and GOF, respectively. Together with a rescue of nocodazole-induced microtubule destabilization by Hmnr GOF, these results argue that Hmnr regulates microtubule stability. Highlighted by the altered movement of a plus-end-associated protein, Hmnr also has an effect on the dynamic nature of microtubules. The authors present evidence suggesting that the nucleation frequency of neuronal microtubules depends on Hmnr's ability to recruit the microtubule nucleator Tpx2. The authors discuss how branching may be regulated by Hmnr-mediated microtubule dynamics and speculate about the physiological significance of altered neuronal morphogenesis. Together, their work adds novel insight into MAP-mediated regulation of microtubules as a prerequisite for neuronal morphogenesis.

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Reviewer #2 (Public Review):

The mechanism of microtubule formation, stabilization, and organization in neurites is important for neuronal function. In this manuscript, the authors examine the phenotype of neurons following alteration in the level of the protein HMMR, a microtubule-associated protein with established roles in mitosis. Neurite morphology is measured as well as microtubule stability and dynamic parameters using standard assays. A binding partner of HMMR, TPX2, is localized. The results support a role for HMMR in microtubule stabilization in neurons.

The results show that HMMR is distributed as puncta on neurons using standard immunofluorescence and PLA. Depletion of HMMR reduced neurite length and extent of branching; reduced post-translational acetylation of neurite microtubules. Conversely,

overexpression of HMMR increased resistance to nocodazole. The parameters of microtubule dynamics were also impacted by reduction or overexpression of HMMR. The authors discuss the possibility HMMR regulates neurite morphological changes via regulation of microtubule nucleation and dynamics.

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Author response:

The following is the authors' response to the original reviews.

eLife assessment

In their valuable study, Chen et al. aim to define the neuronal role of HMMR, a microtubule-associated protein typically associated with cell division. Their findings suggest that HMMR is necessary for proper neuronal morphology and the generation of polymerizing microtubules within neurites, potentially by promoting the function of TPX2. While the study is recognized as a first step in deciphering the influence of HMMR on microtubule organization in neurons, reviewers note the current work has important gaps and would benefit from further exploration of the mechanism of microtubule stability by HMMR, the link between HMMR-mediated microtubule generation and morphogenesis, and the physiological implications of disrupting HMMR during neuronal morphogenesis.

Public Reviews:

Reviewer #1 (Public Review):

The microtubule cytoskeleton is essential for basic cell functions, enabling intracellular transport, and establishment of cell polarity and motility. Microtubule-associated proteins (MAPs) contribute to the regulation of microtubule dynamics and stability - mechanisms that are specifically important for the development and physiological function of neurons. Here, the authors aimed to elucidate the neuronal function of the MAP Hmmr, which they had previously identified in a quantitative study of the proteome associated with neuronal microtubules.

The authors conduct well-controlled experiments to demonstrate the localization of endogenous as well as exogenous Hmmr on microtubules within the soma as well as all neurites of hippocampal neurons. Functional analysis using gain- and loss-of-function approaches demonstrates that Hmmr levels are crucial for neuronal morphogenesis, as the length of both dendrites and axons decreases upon loss of Hmmr and increases upon Hmmr overexpression. In addition to length alterations, the branching pattern of neurites changes with Hmmr levels. To uncover the mechanism of how Hmmr influences neuronal morphology, the authors follow the lead that Hmmr overexpression induces looped microtubules in the soma, indicative of an increase in microtubule stability. Microtubule acetylation indeed decreases and increases with Hmmr LOF and GOF, respectively. Together with a rescue of nocodazole-induced microtubule destabilization by Hmmr GOF, these results argue that Hmmr regulates microtubule stability. Highlighted by the altered movement of a plus-end-associated protein, Hmmr also has an effect on the dynamic nature of microtubules. The authors present evidence suggesting that the nucleation frequency of neuronal microtubules depends on Hmmr's ability to recruit the microtubule nucleator Tpx2. Together, these data add novel insight into MAP-mediated regulation of microtubules as a prerequisite for neuronal morphogenesis. While the data shown support the author's conclusions, the study also has several weaknesses:

- *The study appears incomplete as the initial proteomics analysis which is referenced as an entry into the study is not presented. This surely is the authors' choice, however, without presenting this data set, it would make more sense if the authors first showed the localization of Hmnr on neuronal microtubules and then started with the functional analysis.*

The reviewer suggests moving the Hmnr localization data in front of the loss- and gain-of-function data because we did not present the proteomics data. However, we still believe placing the loss- and gain-of-function data in the beginning is the better arrangement. This is because it allows the audience to see the drastic changes on neuronal morphology when HMMR is depleted or overly abundant. It also provides a better linkage between HMMR's localization on microtubules and its effect on the stability and dynamics of microtubules.

- *Neurite branching is quantified, but the methods used are not consistent (normalized branch density vs. Sholl analysis) and there is no distinction between alterations of branching in dendrites vs. axons. This information should be added as it could prove informative with respect to the physiological function of Hmnr in neurite branching.*

Sholl analysis is considered the gold standard in neurite branching analyses. However, in the knockdown experiment (Figure 1A~1E), HMMR-depleted neurons exhibited extremely short axons (<100 μm) and dendrites (<40 μm). Using Sholl analysis to assess the branching of these Hmnrdepleted neurons became unsuitable. That is why we used normalized branch density (Figure 1E) in the knockdown experiment and Sholl analysis (Figure 1J) in the overexpression experiment.

Regarding the branching difference between axons and dendrites, only axons exhibit branches at 4 DIV. Therefore, the branching analysis focuses on axons rather than on dendrites. We have revised the manuscript to clarify this.

- *The authors show that altered Hmnr levels affect neurite branching and identify an effect on microtubule stability and dynamics as a molecular mechanism. However, how branching correlates with or is regulated by Hmnr-mediated microtubule dynamics is neither addressed experimentally nor discussed by the authors. The physiological significance of altered neuronal morphogenesis also lacks discussion.*
- To discuss how branching correlates with or is regulated by HMMR-mediated microtubule dynamics, we have added the following paragraph into the Discussion section:

“It has been shown that compromising microtubule nucleation in neurons by SSNA1 mutant overexpression prevents proper axon branching (Basnet et al., 2018). Additionally, dendritic branching in *Drosophila* sensory neurons depends on the orientation of microtubule nucleation. Nucleation that results in an anterograde microtubule growth leads to increased branching, while nucleation that results in a retrograde microtubule growth leads to decreased branching (Yalgin et al., 2015). These results demonstrate the importance of microtubule nucleation on neurite branching. It is conceivable that overexpressing a microtubule nucleation promoting protein such as HMMR results in an increase of branching complexity.”

- In terms of discussing the physiological significance of altered neuronal morphogenesis. We have added the following paragraph to the Discussion section:

“Neurons are the communication units of the nervous system. The formation of their intricate shape is therefore crucial for the physiological function. Alterations in neuronal morphogenesis have a profound impact on how nerve cells communicate, leading to a variety of physiological consequences. These consequences include impaired neural circuit formation and function, compromised signal transmission between neurons, as well as altered anatomical structure of the CNS. Depending on the specific type and location of the morphogenetically altered neurons, the physiological consequences can include neurological disorders such as autism spectrum disorder (Berkel et al., 2012) and schizophrenia (Goo et al., 2023), as well as learning and memory deficits (Winkle et al., 2016). However, due to the involvement of HMMR on mitosis, most HMMR mutations are associated with familial cancers (based on ClinVar data).”

- *Multiple times, the manuscript lacks a rationale for an experimental approach, choice of cell type, time points, regions of interest, etc. Also, a meaningful description of the methods and for how data were analyzed is missing, making the paper hard to read for someone not directly from the field.*

We understand the reviewer’s comments regarding the lack of rationale for choosing the experimental approach, choice of cell type, time points, regions of interest, etc. As a result, we have added the rationales where appropriate to help readers from other fields to better understand the choice of cell type, time points, regions of interest, etc. A brief explanation is shown below:

- Approach and timing: We employed both electroporation (immediate but milder expression) and lipofectamine transfection (delayed but stronger expression). We prioritized knocking down HMMR early in development, so electroporation was used. For overexpression experiments, we chose lipofectamine which allows high protein expression level to be achieved.
- Cell selection: Hippocampal neurons were chosen in experiments that involve morphological quantification due to their homogeneous morphology. On the other hand, cortical neurons were selected in experiments that require large amounts of neurons and/or experiments where we want to demonstrate the universality of a proposed hypothesis.
- Regions of interest (ROIs): In our previous publication (Chen et al., 2017), it was discovered that a significant reduction of EB3 emanation frequency can be detected at the tip and the base of the neurite but not in the middle of the neurite in TPX2-depleted neurons. The reason for this difference is due to the presence of GTP-bound Ran GTPase (RanGTP) at the tip and the base of the neurite. Since RanGTP has also been shown to regulate the interaction between HMMR and TPX2 in the cell-free system (Scrofani et al., 2015), it is possible that the same phenomenon can be observed in HMMR-depleted neurons. This is why we examined those 3 ROIs in Figure 4.

Reviewer #2 (Public Review):

The mechanism of microtubule formation, stabilization, and organization in neurites is important for neuronal function. In this manuscript, the authors examine the phenotype of neurons following alteration in the level of the protein HMMR, a microtubule-associated protein with established roles in mitosis. Neurite morphology is measured as

well as microtubule stability and dynamic parameters using standard assays. A binding partner of HMMR, TPX2, is localized. The results support a role for HMMR in neurons.

The work presented in this manuscript seeks to determine if a MAP called HMMR contributes to microtubule dynamics in neurons. Several steps, including validation of the RNAi, additional statistical analysis, use of cells at the same age in culture, and better documentation in figures, would increase the impact of the work.

In many places, the data can be improved which might make the story more convincing. As presented, the results show that HMMR is distributed as puncta on neurons with data coming from a single HMMR antibody, and some background staining that was not discussed. In the discussion the authors state that HMMR impacts microtubule stability, which was evaluated by the presence of post-translational modification and resistance to nocodazole; the data are suggestive but not entirely convincing. The discussion also states that HMMR increases the “amount” of growing microtubules which was measured as the frequency of comet appearance. The authors did not comment on how the number of growing microtubules results in the observed morphological changes.

We actually tested several HMMR antibodies, including E-19 (Santa Cruz, sc-16170), EPR4054 (Abcam, ab124729), and a variety of antibodies provided by Prof. Eva Turley. E-19 performed the best in immunofluorescence (IF) staining and knockdown validation. The other antibodies either failed to detect HMMR in IF staining or generate excessive background signal. We understand that the final images are produced using a single antibody. But since we meticulously validated this antibody and that the localization of overexpressed HMMR is consistent with the endogenous HMMR, we are very confident about our data generated using this single antibody.

We have added the following paragraph in the Discussion section to elucidate how the number of growing microtubules result in the observed morphological changes such as an increase of axon branches:

“It has been shown that compromising microtubule nucleation in neurons by SSNA1 mutant overexpression prevents proper axon branching (Basnet et al., 2018). Additionally, dendritic branching in *Drosophila* sensory neurons depends on the orientation of microtubule nucleation. Nucleation that results in an anterograde microtubule growth leads to increased branching, while nucleation that results in a retrograde microtubule growth leads to decreased branching (Yalgin et al., 2015). These results demonstrate the importance of microtubule nucleation on neurite branching. It is conceivable that overexpressing a microtubule nucleation promoting protein such as HMMR results in an increase of branching complexity.

Reviewer #1 (Recommendations for The Authors):

(1) The manuscript jumps extensively between main figures and supplementary figures. Please check whether parts of the supplement could be moved to the main figures.

We understand the frustration of moving back and forth between the main figures and supplementary figures. After examining the manuscript, we decided to combine Figure 2A with Figure S3.

(2) In Figure 1, total neurite length between days 3 and 4 DIV does not appear to change - can this be true?

Please check or else explain.

We carefully re-examined our raw data and found out the total neurite length of 4 DIV hippocampal neurons expressing non-targeting shRNA (Figure 1B) and that of 3 DIV hippocampal neurons expressing AcGFP (Figure 1G) are indeed very similar. The explanation is that the 3 DIV hippocampal neurons used for Figure 1G was cultured in low-density and in the presence of cortical neuron-conditioned neurobasal medium (as written in Methods, Neuron culture and transfection section). The low-density culture with minimal overlapping neurites allowed us to better quantify total neurite length, because neurons expressing AcGFP-mHMMR sprouted long and highly branched axons. However, the addition of cortical neuron-conditioned neurobasal medium promoted neurite elongation. This is the reason why the total neurite length of 4 DIV hippocampal neurons expressing non-targeting shRNA (Figure 1B) and that of 3 DIV hippocampal neurons expressing AcGFP (Figure 1G) is similar.

(3) Groen et al. have shown that Hmmr also bundles microtubules, a mechanism that surely is important for neuronal microtubules. Please discuss.

We thank the reviewer for pointing out that HMMR also bundles microtubules and have added this to our revised Discussion section:

“It has been shown that the *Xenopus* HMMR homolog XRHAMM bundles microtubules in vitro (Groen et al., 2004). In addition, deleting proteins which promote microtubule bundling (e.g., doublecortin knockout, MAP1B/MAP2 double knockout) leads to impaired neurite outgrowth (Bielas et al., 2007; Teng et al., 2001). These observations are consistent with our data that overexpressing HMMR leads to the increased axon and dendrite outgrowth, while depleting it results in the opposite phenotype (Figure 1).”

(4) Please explain why in Figure 4, cortical neurons were chosen for analysis and why and how the three different ROIs were picked.

To answer the question why we chose cortical neurons for the analyses in Figure 4, it will be important to explain why we used hippocampal neurons for other figures. Primary hippocampal neurons have a high homogeneity in terms of their morphology. This uniform morphology allows more consistent morphological quantification. Figure 4, however, does not involve morphological quantification. We are more confident to conclude that HMMR regulates microtubule dynamics if this effect can be detected in the relatively heterogeneous cortical neurons. These are the reasons why we chose to analyze cortical neurons in Figure 4.

In our previous publication (Chen et al., 2017), it was discovered that a significant reduction of EB3 emanation frequency can be detected at the tip and the base of the neurite but not in the middle of the neurite in TPX2-depleted neurons. The reason for this difference is due to the presence of GTP-bound Ran GTPase (RanGTP) at the tip of the neurite and in the soma. Since RanGTP has also been shown to regulate the interaction between HMMR and TPX2 in the cell-free system (Scrofani et al., 2015), it is possible that the same phenomenon can be observed in HMMR-depleted neurons. This was why we examined those 3 ROIs in Figure 4.

(5) Microtubule looping has been shown to occur in regions prior to branch formation (e.g. Dent et al. 2004). As the authors identify increased looping upon Hmmr GOF, this should be discussed.

We thank the reviewer for pointing out that microtubule looping occurs in regions of branch formation and have added this to our revised discussion:

“It is worth noting that the elevated level of HMMR increases the branching density of axons (Figure 1J) and promotes the formation of looped microtubules (Figure 3A). This is consistent

with the observations that looped microtubules are often detected in regions of axon branch formation (Dent et al., 1999; Dent and Kalil, 2001; Purro et al., 2008).”

Reviewer #2 (Recommendations for The Authors):

(1) The work seeks to gain insight into microtubule behavior in neurons, an important issue.

(2) Several steps, including validation of the RNAi, additional statistical analysis, use of cells at the same age in culture, and better documentation in figures, would increase the impact of the work.

(3) Figure 1 documents the results of experiments in which the HMMR protein was depleted using shRNA. A western blot of cell extracts from control and depleted cells is needed to verify that the protein level is reduced; alternatively, documentation of the reduction in RNA levels in treated cells could be provided. Neurite, axon, and dendrite length and branch density are measured. The neurite length is in microns, and the axon length is normalized to 100% of the non-treated cells. Please use the same for measures for easier comparison. Looking at the images in Figure 1, the length of the dendrites does not look different in the examples shown, whereas the axon appears shorter. This impression is not supported by the quantification. Are representative images shown? Additionally, the authors should report the values for each replicate of the experiment and compare the three averages rather than comparison of lengths from all measurements. A related issue is that the dendrites do not look longer in panel F, following overexpression of HMMR. For examples of using averages of replicates see: <https://pubmed.ncbi.nlm.nih.gov/32346721/>

The reviewer mentioned that Western blot of cell extracts or RNA quantification from control and depleted cells are needed to verify that the protein level is reduced.

Unfortunately, these assays are extremely difficult to perform in primary neurons due to the low transfection efficiency. We believe that the consistent knockdown phenotype from 3 different shRNA sequences (Figure 1A-D) and the immunofluorescence staining in depleted primary neurons (Figure S2) are sufficient to confirm that HMMR level is reduced.

We revised Figure 1C, 1D, 1H, 1I so that axon and dendrite lengths are all in micron.

We selected another image for the non-targeting control in Figure 1A to better demonstrate the reduction of dendrite length when HMMR is knocked down.

We thank the reviewer for the suggestion of comparing the three average values rather than comparing all measurements. We have performed statistical analyses for all our data using the average values and revised the graphs accordingly. While the P-values changed, our conclusions remain the same.

We thank the reviewer for pointing out this discrepancy and have selected another image of the AcGFP control for Figure 1F to better demonstrate the increase of dendrite length when HMMR is overexpressed.

(4) Given the changes in neurite morphology, the authors examine the localization of endogenous and overexpressed. The supplemental figures (see S2 and S3) show evidence that HMMR is present in a punctate pattern by conventional immunofluorescence. This is reasonable evidence that the protein is in a linear pattern along cytoskeletal microtubules and that the signal is present in puncta. Please move this to the main text, perhaps replacing Figure 2A, which is low magnification and very hard to see the HMMR staining. Additionally, the level of overexpression of HMMR is not mentioned. Please address this; were cells with similar levels of overexpression selected? Did the result

depend on the overexpression? A related issue is the DIV for the cells - some are examined earlier and some at later times; does this impact the results? Please provide information or perform experiments with consistent timing. For the immunofluorescence, were multiple antibodies tried to see if the result was the same with each? Were different fixations, in addition to methanol, utilized?

We have replaced Figure 2A with Figure S3 based on the reviewer's suggestion.

In the HMMR overexpression experiments, we used HMMR antibody and immunofluorescence staining to confirm that the overexpression is achieved. However, we did not quantify to what extent HMMR was overexpressed.

We performed all the depletion experiments on 4 DIV to maximize knockdown efficiency and performed all the overexpression experiments on 3 DIV to prevent excessive axon fasciculation. Nonetheless, we examined the effect of HMMR depletion on neuronal morphology on 3 DIV. The trend of reduced total neurite length, axon length, and dendrite length can be observed, but no statistical significance can be detected. We also examined the effect of HMMR overexpression on neuronal morphology on 4 DIV and did observe an increase of total neurite length, axon length, and dendrite length. But the overlapping and bundled axons made reliable quantification extremely difficult.

We actually tested multiple HMMR antibodies, such as E-19 (Santa Cruz, sc-16170), EPR4054 (Abcam, ab124729), and a variety of antibodies provided by Prof. Eva Turley. E19 performed the best in immunofluorescence (IF) staining and knockdown validation. The other antibodies either failed to detect HMMR in IF staining or generate excessive background signal. We also tested various fixation methods, including 37°C formaldehyde fixation, -20°C methanol fixation, 37°C formaldehyde followed by -20°C methanol fixation. All fixation methods generated similar IF staining pattern using the E-19 antibody, but 3.7% formaldehyde fixation produced the highest signal.

(5) In Figure 2 C it is hard to see DAPI fluorescence. Are the white areas in the merge with bright cell nuclei? Is Figure 2C control or overexpressing cells? If this is endogenous, is there less signal in PLA compared with S4, which was in culture longer and is overexpressed prior to using PLA for detection?

The white areas in Figure 2C the reviewer mentioned are not cell nuclei, they are actually bubbles formed within the mounting medium.

HMMR detected in Figure 2C is endogenous. We did not quantitatively compare the PLA signals in Figure 2C and those in Figure S4. This is because the PLA signals in Figure 2C are generated using anti-HMMR (to detect endogenous HMMR) and anti- β -III-tubulin antibodies while those in Figure S4 are generated using anti-AcGFP (to detect overexpressed AcGFP-mHMMR) and anti- β -III-tubulin antibodies. Since the affinity of the two antibodies (i.e., anti-HMMR and anti-AcGFP) toward their antigens is different, comparing the PLA signals is not informative.

(6) The images of the endogenous HMMR (Fig S3) and the PLA with tubulin and HMMR antibodies are not the same (2C). The "dots" in PLA are widely separated; gauging from the marker bar length of 50 μ m, the small clusters of dots are about 10 μ m apart. In Figure S3, the puncta are much more closely spaced, appearing almost in a linear fashion along the microtubules. Enlarging the PLA image shows that each dot is very small - just a few pixels - please provide additional explanation including the minimal detection limit for the method, and why the images differ. If the standard immunofluorescence signal was enhanced, for example with the use of two secondaries, what is observed? Is the distribution of HMMR similar for both dendrites and axons?

Microtubule polarity differs in these locations, so greater attention to this point seems of interest. There is a significant amount of punctate HMMR in the cytoplasm (or outside the cytoplasm?) in Figure S5; this is concerning. Please outline the cell edge for ease of visualization. What is the distribution of HMMR in a cell that has been treated with cold and/or nocodazole to disassemble the microtubules? is the signal lost?

The reasons images of the endogenous HMMR (Figure S3) and the PLA with tubulin and HMMR antibodies (Figure 2C) differ are due to the following reasons. o PLA utilizes two primary antibodies to target two different epitopes on HMMR and β III-tubulin. It is conceivable that not every anti-HMMR antibody has the correct orientation and/or proximity (<40 nm) toward the anti- β -III-tubulin antibody to enable DNA amplification. This results in the shortage of PLA puncta compared to immunofluorescence signals.

- The creator of PLA has pointed out that in situ PLA is a method based upon equilibrium reactions and several enzymatic steps. Therefore, only a fraction of the inter-acting molecules is detected (Weibrecht et al., 2010).

We have not used signal enhancing immunofluorescence staining methods [e.g., using tertiary antibodies or tyramide signal amplification (TSA)] to detect HMMR. This is mainly because HMMR signal is strong enough to be detected using standard immunofluorescence staining.

Regarding the question “Is the distribution of HMMR similar for both dendrites and axons?” The reviewer raised a very important issue about the polarity difference of microtubules in axons (uniform) and dendrites (mixed). We were aware of such issue and very carefully examined the distribution and signal intensity of HMMR in axons vs dendrites. However, no differences were detected.

The reviewer mentioned that “there is a significant amount of punctate HMMR in the cytoplasm (or outside the cytoplasm?) in Figure S5; this is concerning. Please outline the cell edge for ease of visualization.” Instead of outlining the cell edge, we have selected another image to facilitate the visualization of HMMR signals. There are indeed HMMR signals outside the cell. However, these outside signals are usually weaker and smaller in size compared to those inside the cell.

After the examination of neurons expressing AcGFP-mHMMR with or without 100 nM nocodazole treatment, we did not notice any difference of AcGFP-mHMMR in distribution. We did not examine the distribution and signal intensity of the endogenous HMMR.

(7) To determine if HMMR alters microtubule stability, the authors examine the distribution of acetylated tubulin and resistance to nocodazole-induced microtubule disassembly. In Figure 3 please show immunofluorescence images of the acetylated tubulin staining, not just the ratio images; the color is not obviously different in the various panels shown. For statistical analysis, see the comment above for Figure 1. For the nocodazole experiment, a similar change in neurite length following drug treatment was observed (Figure 3H), for the experimental and control, even though the starting length was greater in the overexpressing cells. Please consider the possibility that in both cases the microtubules are only partially resistant to nocodazole and that HMMR is not changing the fraction of microtubules that are sensitive to the drug. The cells were treated at 3 DIV; the authors note that more stable microtubules accumulate with time; how does time in culture impact stability? Often, acute treatment with a high concentration of nocodazole is used to assay microtubule stability; here the authors used a low (nM) concentration for 2 days (chronic). Why not use a higher concentration (1-10 μ M) for a shorter incubation? The data show that overexpression of HMMR results in

curved, buckled microtubules are these microtubules more acetylated and/or retained after nocodazole treatment?

The reviewer suggested that we show immunofluorescence images of the acetylated tubulin staining, not just the ratio images. But we still believe showing the ratio images is the better approach. This is because the microtubules density can be different from neuron to neuron. Showing acetylated tubulin may provide a false impression when the overall microtubule density is higher or lower in a particular neuron. We realized that “16 colors” pseudo-color scheme has the cyan color at the lower intensity which can sometimes be confused with the white color at the higher intensity. Therefore, we changed the pseudocolor from “16 colors” to “fire” for Figure 3B and 3E to better visualize these images so that they appear more consistent with the quantitative data.

The reviewer raised a very good question regarding the possibility that HMMR is not changing the fraction of microtubules that are sensitive to nocodazole. We re-conducted the same experiment and used a series of different nocodazole concentrations. While the addition of nocodazole causes a concentration-dependent reduction of total neurite length in both AcGFP and AcGFP-mHMMR expressing neurons, there are subtle differences in the susceptibility of neurite length to the concentration of nocodazole. 1) 10 nM nocodazole treatment causes a significant reduction of neurite length in AcGFP expressing neurons, but not in AcGFP-mHMMR expressing neurons. This result indicates that AcGFP-mHMMR expression increases the tolerance of neurite elongation toward 10 nM nocodazole treatment. 2) 50 nM and 100 nM nocodazole treatment exhibits no statistical significance in AcGFP expressing neurons, suggesting that 50 nM nocodazole has reached maximal effectiveness. In AcGFP-mHMMR expressing neurons, 100 nM nocodazole further reduces the neurite length compared to the 50 nM group. These results argue against the possibility that HMMR does not change the fraction of microtubules that are sensitive to nocodazole. We have revised Figure 3H accordingly.

The reviewer asked why we did not use the acute nocodazole treatment (μM concentration) to assess the effect of Hmmr on microtubule stability. This is because we used the neurite length as an indicator for microtubule stability. That is why the chronic treatment was chosen to produce a more detectable effect on neurite length.

The reviewer asked whether the looped microtubules caused by HMMR overexpression are more acetylated and/or nocodazole resistant. While we do not have direct evidence to answer the reviewer’s question, we can deduce the answer from our observations. We noticed that looped microtubules are only present when HMMR is highly expressed (i.e., using lipofection to introduce HMMR-expressing plasmid) but not when HMMR is mildly expressed (i.e., using electroporation to introduce HMMR-expressing plasmid). From these observations, we can conclude that HMMR is more abundantly present on looped microtubules. Since HMMR overexpression leads to higher microtubule acetylation (Figure 3E), looped microtubules which contains more HMMR are most likely to be more acetylated.

(8) An additional measure of microtubule dynamics is to measure the growth of microtubules using a live cell marker for microtubule plus ends. Such experiments were performed, using tagged EB3. The images are rather fuzzy. Parameters of microtubule dynamics were measured at three locations - is there data that the authors can cite about any differences in dynamics in control cells at these locations? They look very similar, so it is not clear why the different locations were used. It is not possible to learn much from the kymographs which look similar for all panels; I would remove these unless they can be changed or labeled to help the reader. Data is presented for three shRNA reagents. No data are presented to document the extent to which the protein is depleted with these reagents. This should be fixed. Alternatively, an RNAi pool could be utilized. Is there a control for off-target effects? For the analysis were all the comets used

to generate the average values? What about a comparison of the average of each trial - not each comet?

In our previous publication (Chen et al., 2017), it was discovered that a significant reduction of EB3 emanation frequency can be detected at the tip and the base of the neurite but not in the middle of the neurite in TPX2-depleted neurons. The reason for this difference is due to the presence of RanGTP at the tip and the base of the neurite. Since RanGTP has also been shown to regulate the interaction between HMMR and TPX2 in the cell-free system (Scrofani et al., 2015), it is possible that the same phenomenon can be observed in HMMR-depleted neurons. This is why we examined those 3 ROIs in Figure 4.

We notice that photobleaching causes the EB3-mCherry signal to diminish at later time points, which made it difficult to observe the differences amongst kymographs. In the revised Figure 4B and 4D, we removed the second half of all the kymographs to make the differences more obvious.

The reviewer mentioned that there are no data documenting the extent to which the protein is depleted with the shRNAs. These data are shown in Figure S2, in which we quantified the HMMR protein level in the soma and along the neurite in neurons expressing different shRNA molecules.

The reviewer asked whether there is a control for off-target effects. The answer is yes. We performed the rescue experiment to control for off-target effects, which is shown in Figure S1.

We revised Figure 4 so that the dynamic properties of EB3 are quantified using the average of each experimental repetition.

(9) In a final experiment, the authors examine the distribution of TPX2, a binding partner of HMMR. Include a standard immunofluorescence in addition to PLA to illustrate the distribution of TPX2. The quantification used was the inter puncta distance; please quantify the signal in control and treated cells.

The reviewer asked us to include a standard immunofluorescence staining to illustrate the distribution of TPX2. We have done that in our previous publication (Chen et al., 2017) and TPX2 localizes primarily to the centrosome (<https://www.nature.com/articles/srep42297/figures/2>). In order to enhance the weak signal of TPX2 along the neurite, we actually needed to use PLA in that publication (<https://www.nature.com/articles/srep42297/figures/3>).

Proximity ligation assay (PLA) generates fluorescent signals based on a local enzymatic reaction which catalyzes the amplification of a specific DNA sequence that can then be detected using a red fluorescent probe. Because this enzymatic reaction is not linear, the amount of amplified DNA nor the intensity of the fluorescence does not correlate with the strength of the interaction (Soderberg et al., 2006). As a result, quantification of PLA is typically done by counting the number of fluorescent puncta per unit area or by calculating the area containing fluorescent signal (not signal intensity) per unit area in the case that PLA signals are too strong and coalesced. That is why our quantification is based on the distance between PLA fluorescent puncta, not the fluorescent signal intensity.

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